

A Higher- Q Scale Audit of a Fixed Prime-Shell Tilt

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Abstract

TPC-365 found that a beta=2 prime-shell tilt survived a response-blind finite holdout through $Q = 512$. We now freeze beta=2 and attack scale on a new geometry-selected panel, using the five-anchor ladder $Q \in \{512, 1024, 2048, 4096, 8192\}$. Forty-one candidate origins are scored only by unsigned weighted square geometry on 256-point pilots; the declared greedy rule selects (623071, 631360, 629211). The complete replay has 480 rows, with two betas, two counts, two kernel exponents, four fixed sign laws, and all five shell anchors. Beta=2 has zero spectral-cap and zero Schur-cap violations in all 240 rows, with maximum normalized spectrum 0.62448287758976528 and maximum normalized Schur value 0.65368278287004711. The beta=0 control has 60 violations of each cap. This is a finite higher- Q observation, not a shell-uniform operator theorem, a source-valid arithmetic normalization, a Route-A/Route-B pass, or a twin-prime result.

1 Question and scope

TPC-362 located a finite cap failure beginning at $Q = 128$ for the inherited normalization. TPC-364 introduced the explicit shell weight

$$w_{p,\beta} = \left(\frac{p}{Q}\right)^\beta, \quad (1)$$

and TPC-365 showed that the fixed choice beta=2 transferred to a new finite panel through $Q = 512$. The present question is narrower: after beta=2 is frozen, how far does the finite cap observation extend in Q ?

The panel is still geometry-selected. It is therefore useful for mapping a route and locating scale obstructions, but it is not a random independent sample or a uniform-in-origin claim. No source vector, adaptive sign, or arithmetic reassembly is used. The official Session-named Route-A and Route-B evaluator files are absent from this checkout; local Bridge-B is fail-closed evidence only.

2 Finite operator and frozen selection

For $I = [x, x + N - 1] \cap \mathbb{Z}$ and $Q < p \leq 2Q$, define

$$B_p(u, t) = p \frac{66^{2s}}{(66^2 + (u - t)^2)^s} \left(\mathbf{1}_{p|u-t} - \frac{1}{p-1} \right) \mathbf{1}_{u \neq t} \mathbf{1}_{p \nmid u} \mathbf{1}_{p \nmid t}. \quad (2)$$

For integer beta, let

$$A_{\beta,\varepsilon} = \sum_{Q < p \leq 2Q} \varepsilon_p \left(\frac{p}{Q}\right)^\beta B_p, \quad G_{\beta,u} = \sum_{Q < p \leq 2Q} \sum_{t \in I} \left(\left(\frac{p}{Q}\right)^\beta B_p(u, t) \right)^2. \quad (3)$$

The normalized matrix is the finite symmetric congruence

$$A_{\beta,\varepsilon}^\# = D_{G_\beta}^{-1/2} A_{\beta,\varepsilon} D_{G_\beta}^{-1/2}, \quad D_{G_\beta} = \text{diag}(G_{\beta,u}). \quad (4)$$

Table 1: Complete beta census on the higher- Q panel (240 rows per beta).

beta	max spectrum	spectral failures	max Schur	Schur failures
0	1.6419614116	60	1.7182186230	60
2	0.6244828776	0	0.6536827829	0

Table 2: Maximum normalized spectrum at each shell anchor.

Q	beta=0		beta=2	
	max spectrum	failures	max spectrum	failures
512	1.6398880045	12	0.6163019781	0
1024	1.6409838731	12	0.6244828776	0
2048	1.6415483555	12	0.6196009007	0
4096	1.6418250155	12	0.6229914076	0
8192	1.6419614116	12	0.6208083916	0

The geometry is a sum of rational squares. Once it is positive, the only operator bounds used are the elementary finite envelopes

$$\|\cdot\|_2 T \leq \max_u \sum_t |T(u, t)|, \quad \|\cdot\|_2 T \leq \left(\sum_{u,t} |T(u, t)|^2 \right)^{1/2}. \quad (5)$$

Candidate origins are $a_j = 620001 + 307j$, $0 \leq j < 41$. On $[a_j, a_j + 255]$, calculate beta=2 geometry at every declared (Q, s) and score

$$S(a_j) = \max_{Q,s} \frac{\max_u G_{2,u}}{\min_u G_{2,u}}, \quad Q \in \{512, 1024, 2048, 4096, 8192\}, \quad s \in \{1, 2\}. \quad (6)$$

Sort by decreasing score and increasing origin, then greedily retain origins separated by at least 2048. The ordered output is (623071, 631360, 629211). The top score is 2.0552882031131818 at origin 623071, attained at $(Q, s) = (512, 2)$. The signed matrices are evaluated only after this rule is frozen.

3 Higher- Q audit

The protocol uses counts 256, 512, exponents 1, 2, four fixed laws (all-plus, alternating shell index, prime modulo-four character, and half-shell split), and beta 0, 2. Thus it contains $2 \cdot 3 \cdot 2 \cdot 5 \cdot 2 \cdot 4 = 480$ rows, with a true spectrum for every law. The spectral value 0.64 and Schur value 0.83 are inherited finite working caps only.

The beta=0 failures occur at every shell anchor in this declared panel. The beta=2 maximum remains below both finite caps, but it is not monotone: it rises from the TPC-365 value 0.61633188509480319 to 0.62448287758976528, a difference of 0.0081509924949620949. We record this increase rather than assigning a decay or transfer theorem. The minimum effective shell fraction is 0.66944805377549699, so the finite observation is not described as a one-prime truncation.

4 Exact and independent verification

At $Q = 4$, exponent one, the half-open interval [623372, 623385) has shell {5, 7}. Exact rational matrix and geometry digests are retained for beta=0 and beta=2; symmetry and positivity are checked by exact arithmetic.

The forward producer evaluates 480 rows in increasing shell order. An independently written checker rebuilds the sieve, response-blind selection, masks, weights, four laws, geometry, finite envelopes, and all true spectra in reverse shell order. It compares every row's shell, weights, geometry extrema, raw and normalized metrics, row index, phase census, and exact anchor. A separate adversarial checker rejects 23 mutations of protocol, selection, scale limits, phase counts, and claim-firewall fields. Local Bridge-B requires empty stderr and byte-identical stdout in normal and optimized modes.

5 Claim firewall and route decision

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TPC366_GEOMETRY_SELECTION = PROVED_EXACT_FINITE_RESPONSE_BLIND
TPC366_WEIGHTED_GEOMETRY_POSITIVITY = PROVED_EXACT_FINITE
TPC366_FINITE_REPLAY = NUMERICALLY_CERTIFIED_FINITE_480_ROWS
TPC366_HIGHER_Q_LADDER = NUMERICALLY_CERTIFIED_FINITE_SCOPED
TPC366_BETA2_HIGHER_Q_CAP = NUMERICALLY_CERTIFIED_FINITE_SCOPED
TPC366_BETA2_SCALE_UNIFORMITY = OPEN
TPC366_BETA2_ASYMPTOTIC_REPAIR = OPEN
TPC366_NORMALIZATION_SOURCE_VALIDITY = MODELING_CHOICE_OPEN
TPC366_GROWING_OPERATOR_BOUND = OPEN
TPC366_SOURCE_UNIFORM_L2 = OPEN
TPC366_ARITHMETIC_ADVANCE = NO
TPC366_FIXED_POWER_CREDIT = 0
TPC366_FULL_GATE_B = OPEN
TPC366_TWIN_PRIME_RESULT = NONE
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The strongest positive result is a fixed $\beta=2$ cap observation through $Q = 8192$ on the declared finite panel. The strongest obstruction is the nonmonotone finite scale profile together with geometry-based origin selection and the absence of source validity. No Route-A or Route-B gate is closed, no fixed power is credited, and no twin-prime conclusion follows. The next minimal attack is to keep $\beta=2$ fixed while using longer windows and predeclared or unselected origins.

6 Conclusion

TPC-366 extends the finite $\beta=2$ signal by four additional shell scales, through $Q = 8192$, while the $\beta=0$ control continues to fail at every anchor. This is useful route reconnaissance: the immediate obstruction is no longer the tested shell size, but the lack of a uniform statement across windows, origins, and the arithmetic source. The result remains a finite modeling audit. A growing operator bound, source-uniform L^2 , source-valid normalization, reassembly, and the twin-prime endpoint remain open.

TPC366_ARITHMETIC_ADVANCE=NO, TPC366_FIXED_POWER_CREDIT=0, TPC366_FULL_GATE_B=OPEN.